

Rumaisa Habib

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Art by Haider Naveed

Research Overview

My research focuses on understanding the web at a global scale. I combine large-scale measurement and systems building to study how the web works in practice and who it actually serves, especially outside Western contexts. I aim to replace informal intuitions about the web with rigorous evidence that can inform both system design and policy.

Education

Ph.D. Candidate in Computer Science, Stanford University — Expected 2028
M.S. in Computer Science, Stanford University — 2026
B.S. in Computer Science, LUMS — 2023

➔ Senior Thesis: *A Framework for Improving Web Affordability and Inclusiveness*

Publications

[1] "Stop Trying to Make Fetch Happen": Towards a Modality-Aware Understanding of Unnecessary Resources on the Web
Rumaisa Habib, Balaji Balachandran, Nurullah Demir, Zakir Durumeric
ACM Internet Measurement Conference (IMC '26)

[2] On IP Addresses as Identifiers of Internet Users and Services
Rumaisa Habib, Sudheesh Singanamalla, Marwan Fayed, Zakir Durumeric
Research Conference on Communications, Information and Internet Policy (TPRC '26)

[3] Don't Get Caught, Keep Your Onions in a Vault
Humza Ikram*, **Rumaisa Habib***, Muaz Ali*, Zartash Afzal Uzmi (*joint first authors)
New Ideas in Networked Systems (NINeS '26)

[4] Formalizing Dependence of Web Infrastructure
Rumaisa Habib, Kimberly Ruth, Gautam Akiwate, Zakir Durumeric
ACM Conference of the Special Interest Group on Data Communication (SIGCOMM '25)

IRTF Applied Networking Research Prize

[5] Blade: Towards Scalable Source Code Debloating
Muaz Ali, **Rumaisa Habib**, Ashish Gehani, Sazzadur Rahaman, Zartash Afzal Uzmi
IEEE Secure Development Conference (SecDev '23)

[6] A Framework for Improving Web Affordability and Inclusiveness
Rumaisa Habib*, Sarah Tanveer*, Aimen Inam, Haseeb Ahmed, Ayesha Ali, Zartash Afzal Uzmi, Zafar Ayyub Qazi, Ihsan Ayyub Qazi (*joint first authors)
ACM Conference of the Special Interest Group on Data Communication (SIGCOMM '23)

[7] A First Look at Public Service Websites from the Affordability Lens
Rumaisa Habib, Aimen Inam, Ayesha Ali, Ihsan Ayyub Qazi, Zafar Ayyub Qazi
ACM Web Conference (TheWebConf '23)

[8] Rethinking Web for Affordability and Inclusion
Ihsan Ayyub Qazi, Zafar Ayyub Qazi, Ayesha Ali, Muhammad Abdullah, **Rumaisa Habib**
ACM Workshop on Hot Topics in Networks (HotNets '21)

Preprints

Understanding the "Airport" Censorship Circumvention Ecosystem in China
Rumaisa Habib*, Mingshi Wu*, Shiva Shahandeh, Min Ni, Eric Wustrow, Zakir Durumeric (*joint first authors)
arXiv preprint.

Reports & Blog Posts

- **Author**, *The 2025 Web Almanac* (HTTP Archive) — *"Chapter 8: Privacy"* (January 2026) by Rumaisa Habib, Vinod Tiwari, Nurullah Demir [Link]
- **Guest Author**, Internet Society Pulse — *"Characterizing Internet Centralization vs Regionalization"* (December 2025) [Link]

Grants, Fellowships & Honors

- **IRTF Applied Networking Research Prize** (2026)
- **David Cheriton Stanford Graduate Fellowship**, Stanford University — 2023–Present
- **Students as Co-Researchers Grant**, LUMS — 2023
- **Graduated with High Distinction**, LUMS — 2023
- **Merit Scholarship**, LUMS — 2020–2023
- **Dean's Honor List**, LUMS — 2019–2022

Teaching Experience

- **Course Assistant**, Department of Computer Science, Stanford University
  - CS356: Topics in Computer and Network Security (Fall 2026)
- **Teaching Assistant**, Department of Computer Science, LUMS
  - CS678/EE678: Topics in Internet Research (Spring 2023)
  - CS334/EE334: Principles and Techniques of Data Science (Fall 2022)
  - CS3812: Introduction to Blockchain: Technology and Applications (Spring 2022)
  - CS471/EE471/CS573: Computer Networks: Principles & Practices (Fall 2021)

Conference Presentations & Invited Talks

- **"Stop Trying to Make Fetch Happen": Towards a Modality-Aware Understanding of Unnecessary Resources on the Web**
  - IMC '26, October 2026, Karlsruhe, Germany
- **On IP Addresses as Identifiers of Internet Users and Services**
  - TPRC '54, September 2026, Washington DC, USA
- **Formalizing Dependence of Web Infrastructure**
  - Stanford Security Lunch, July 2026, Stanford, CA, USA
  - IRTF Open, IETF-126, July 2026, Vienna, Austria
- **A Framework for Improving Web Affordability and Inclusiveness**
  - IAB BIAS Workshop, January 2024, Virtual
  - UW CHANGE Seminar, November 2023, Virtual, University of Washington, USA
  - SIGCOMM '23, September 2023, Columbia University, NY, USA
- **A First Look at Public Service Websites from the Affordability Lens**
  - TheWebConf '23, May 2023, UT Austin, TX, USA

Service & Professional Affiliations

Departmental & University Service

- **Organizer**, Weekly Security Seminar, Department of Computer Science, Stanford University — 2024–2025
- **IT & Design Director**, LUMS Women in Computing (LWiC) — 2021–2022
- **Research Lead**, Blockchain Research Group, LUMS — Summer 2022

External Review

- ➔ USENIX Security 2026
- ➔ Internet Measurement Conference (IMC) 2024

Press/Interviews

- ➔ **Wall Street Journal** — *"FBI Shuts Down Sprawling China-Linked Hacking Network"* (August 2026) [Link]
- ➔ **PPC Land** — *"Stanford study finds 5% of IP addresses send 55% of all web requests"* (August 2026) [Link]
- ➔ **Tech Times** — *"IETF 126 Closes Out in Vienna: DPI, Fingerprinting, Prize & BGP Security Push"* (July 2026) [Link]

Updated Oct 22, 2026